Further Integration Techniques

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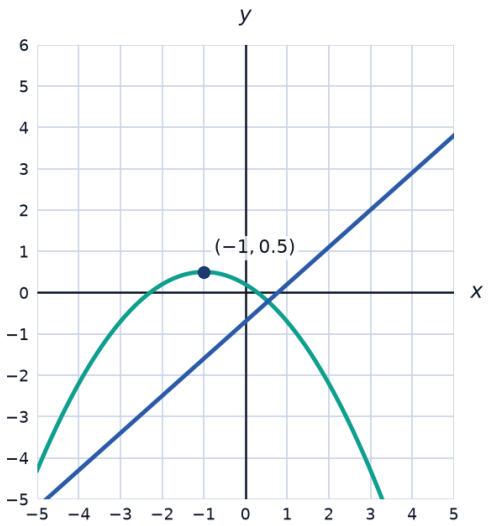
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Integration by substitution

Evaluate $\int 2x(x^2 + 1)^4\,dx$ using the substitution $u = x^2 + 1$.

2 [Foundational] • Target: Integration by parts

Evaluate $\int xe^x\,dx$ using integration by parts.



3 [Foundational] • Target: Repeated integration by parts

Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.

4 [Foundational] • Target: Integration using trigonometric identities

Evaluate $\int \cos^2 x\,dx$ using the identity $\cos^2 x = (1 + \cos 2x)/2$.

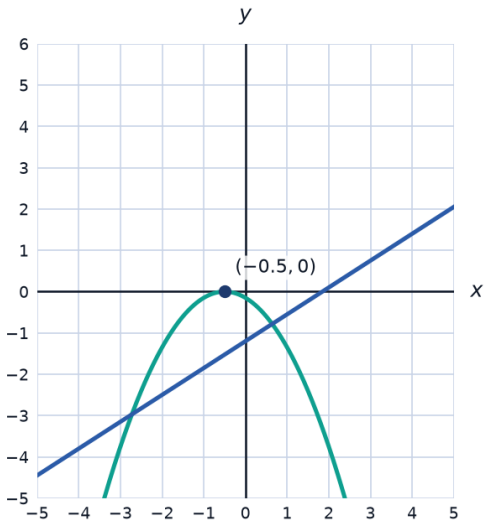
5 [Foundational] • Target: Integration using partial fractions

Evaluate $\int 1/(x^2 - 1)\,dx$ by first expressing the integrand in partial fractions.

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6 [Foundational] • Target: Integration involving inverse trigonometric results

Evaluate $\int 1/(1+x^2)\,dx$, and use it to evaluate $\int_0^1 1/(1+x^2)\,dx$.



7 [Foundational] • Target: Integration by substitution with trigonometric functions

Evaluate $\int \sin^3 x\,dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration using a Weierstrass-style substitution

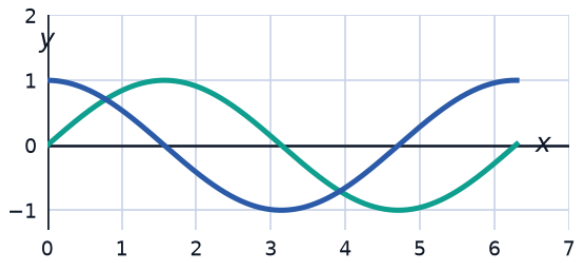
Evaluate $\int 1/(5+4\cos x)\,dx$ using the substitution $t = \tan(x/2)$, given that this substitution transforms $\cos x = (1-t^2)/(1+t^2)$ and $dx = 2/(1+t^2)dt$, leading to the simplified integrand $2/(9+t^2)$.

9 [Intermediate] • Target: Integration by parts with a trigonometric-exponential product

Evaluate $\int e^x \cos x\,dx$, applying integration by parts twice and solving the resulting equation for the original integral.

10 [Intermediate] • Target: Finding the area between two curves using integration

Find the area of the region enclosed between $y = \sin x$ and $y = \cos x$ for $\pi/4 \le x \le 5\pi/4$ (the interval on which $\sin x \ge \cos x$).



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11 [Intermediate] • Target: Finding the area between two curves using integration

A particle's velocity is given by $v(t) = t\cos t$ metres per second. Find the particle's displacement over the interval $0 \le t \le \pi$, using integration by parts.

12 [Intermediate] • Target: Finding the area between two curves using integration

This question has three parts. (a) Use the substitution $u = x^2 + 4$ to find $\int x/(x^2+4)\,dx$. (b) Use integration by parts to find $\int \arctan x\,dx$. (c) Hence, using the result of part (b) and the standard result $\int 1/(1+x^2)dx = \arctan x + C$, evaluate $\int_0^1 \arctan x\,dx$, giving an exact answer in terms of π and $\ln$.

13 [Intermediate] • Target: Reduction formulas

Given the reduction formula $I_n = \int x^n e^x\,dx = x^n e^x - nI_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x\,dx$.

14 [Intermediate] • Target: Volume of revolution using integration

The region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is rotated 360° about the x-axis. Find the volume of the resulting solid, using $V = \pi\int_0^4 y^2\,dx$.

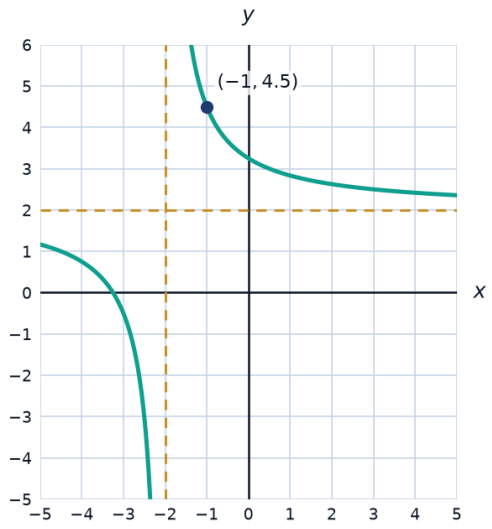
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Integration using the sine double-angle identity • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x-3}{x^2-2x-3}\,dx$

16 [Challenging] • Target: Volume of revolution using integration • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Decompose the rational expression $R(x) = \frac{7x-1}{x^2-x-6}$ into partial fractions using the Heaviside cover-up method.



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17 [Challenging] • Target: Substitution with a shifting definite integral bound • Cross-Lesson: 4.02 The Poisson Distribution

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

18 [Challenging] • Target: Choosing an appropriate integration technique • Cross-Lesson: 5.02 Differential Equations

Level 1 · Foundations Evaluate the indefinite integral $\int \frac{\cos(\ln x)}{x}\,dx$ for $x > 0$.

19 [Challenging] • Target: Integration by substitution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Integration by parts • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Choosing an appropriate integration technique

Which integration technique is most appropriate for evaluating $\int x^3 \sin x\,dx$?

a. Integration by parts, applied three times (reducing the power of x each time)

b. Substitution $u = \sin x$

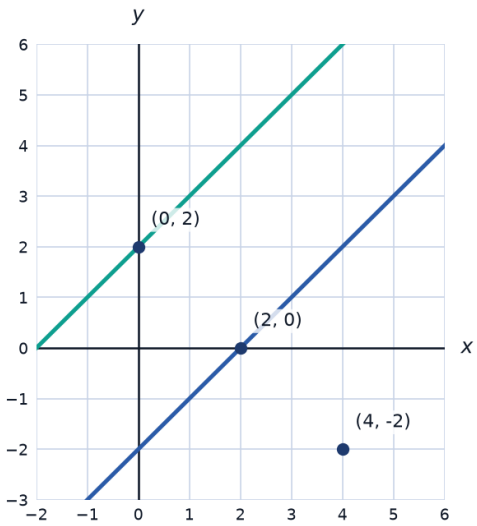
c. Partial fractions

d. The double-angle identity for $\sin$

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22 [Foundational] • Target: Integration by parts

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?



a. 17

b. 7

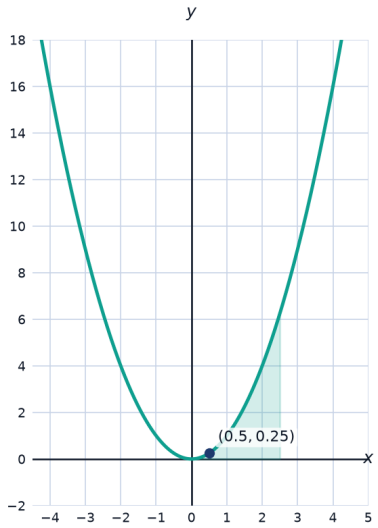
c. 2.4

d. -7

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23 [Foundational] • Target: Repeated integration by parts

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?



a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

24 [Foundational] • Target: Integration using trigonometric identities

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_{a}^{b}(5 - x)\sqrt{x}\,dx$. Find the value of k.

a. 2

b. 4/3

c. 8/3

d. 16/3

25 [Foundational] • Target: Integration using partial fractions

The integral $\int_{0}^{1} e^{-x^2}dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$

c. $1 + 1/3 \approx 1.333$

d. 0

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26 [Foundational] • Target: Evaluating a definite integral

Integration by parts chooses u such that:

a. u is simple to integrate

b. du/dx is simpler than u

c. dv is the whole integrand

d. u contains e^x always

27 [Foundational] • Target: Integration involving inverse trigonometric results

$\int_a^b [f(x) - g(x)]dx$ gives area between curves when:

a. $f > g$ and $f < g$ alternates

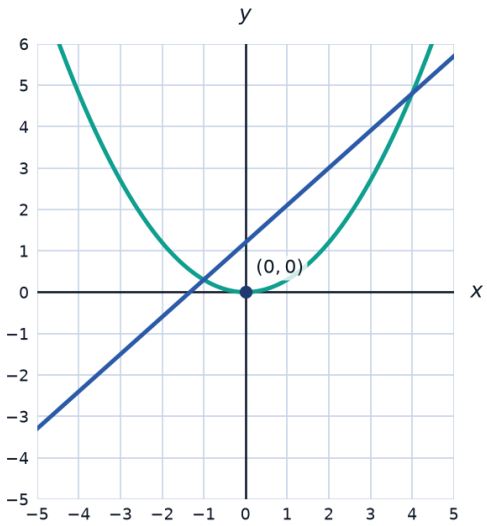
b. $f > 0$ and $g < 0$

c. $f(x) \geq g(x)$ on $[a, b]$

d. always

28 [Foundational] • Target: Integration by substitution with trigonometric functions

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.



a. $\frac{5\pi}{16}$

b. $\frac{5\pi}{32}$

c. $\frac{3\pi}{16}$

d. $\frac{1}{6}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Integration by parts with a logarithmic integrand

Find the volume when the area between $y = \sec x$ and $y = \cos x$ (from $-\pi/4$ to $\pi/4$) is rotated about the x-axis:

a. $\pi^2/4$

b. $2\pi(\pi/4 - 1/2)$

c. $\pi(2 - \pi/2)$

d. $\pi(\sqrt{2} - 1)$

30 [Intermediate] • Target: Integration using a Weierstrass-style substitution

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π

d. $\sqrt{\pi}$

31 [Intermediate] • Target: Integration by parts with a trigonometric-exponential product

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$

d. $x = \log_{100}(7)$

32 [Intermediate] • Target: Finding the area between two curves using integration

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

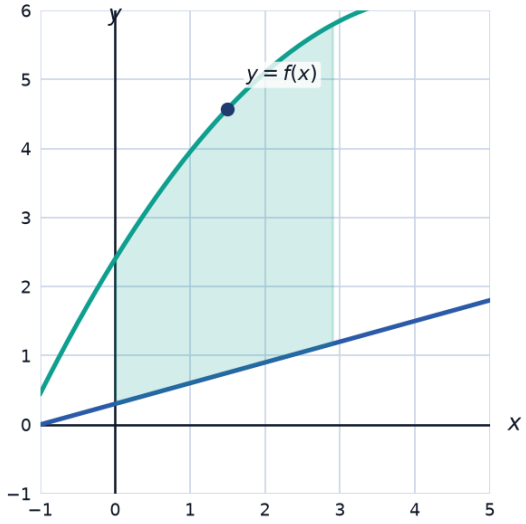
c. $\frac{3}{5}$

d. $\frac{7}{10}$

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33 [Intermediate] • Target: Reduction formulas

Surface area of sphere radius R using surface area of revolution:



a. $4\pi R^2$

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

34 [Intermediate] • Target: Integration by substitution with a linear inner function

If $\sin\theta = \frac{3}{5}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{37}{50}$

b. $\frac{16}{25}$

c. $\frac{4}{5}$

d. $\frac{9}{25}$

35 [Intermediate] • Target: Integration using the sine double-angle identity

Partial fractions: $\frac{2x+3}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$. Find A and B:

a. $A = 2, B = 1$

b. $A = 1, B = 2$

c. $A = 2, B = 3$

d. $A = 3, B = 2$

36 [Intermediate] • Target: Volume of revolution using integration

Volume integral: using spherical coordinates, $dV =$

a. $dr\,d\theta\,d\phi$

b. $r^2\sin\phi\,dr\,d\theta\,d\phi$

c. $r\sin\theta\,dr\,d\theta\,d\phi$

d. $r^2\,dr\,d\theta\,d\phi$

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37 [Intermediate] • Target: Substitution with a shifting definite integral bound

Weierstrass substitution $t = \tan(x/2)$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$. Converts rational trig integrands to:

a. trig of t

b. rational functions of t

c. exponentials of t

d. polynomials

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing an appropriate integration technique • Cross-Lesson: 5.02 Differential Equations

$\int \frac{1}{x(x-1)}\,dx$ by partial fractions:

a. $\ln|x| - \ln|x-1| + C$

b. $\ln|x(x-1)| + C$

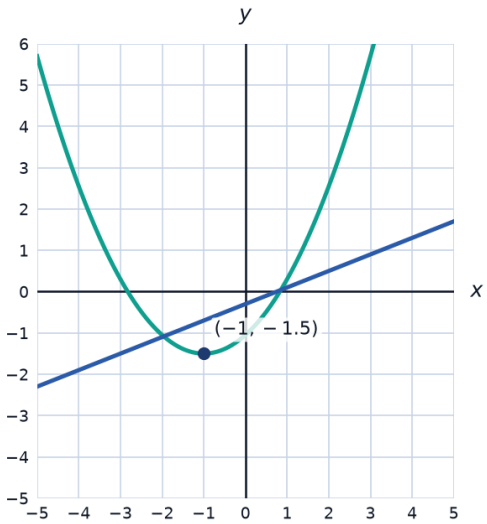
c. $\frac{1}{x-1/2} + C$

d. $\ln|x-1| - \ln|x| + C$

39 [Challenging] • Target: Integration by substitution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Integration of inverse functions (Laisant-Hermite): $\int_a^b f^{-1}(y)dy = bf^{-1}(b) - af^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f(x)dx$.

Example for $f = e^x$:



a. gives nothing useful

b. yields: $\ln b - \ln a$

c. gives $e^b - e^a$

d. yields: $b\ln b - a\ln a - (b-a)$

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40 [Challenging] • Target: Integration by parts • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

$\int \frac{1}{x^3 - x}\,dx$: partial fractions decomposition is:

a. $\frac{A}{x^2+1} + \frac{B}{x-1}$

b. $\frac{A}{x^3} + \frac{B}{x}$

c. $\frac{Ax+B}{x^2} + \frac{C}{x-1}$

d. $\frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$

41 [Challenging] • Target: Repeated integration by parts • Cross-Lesson: 6.01 Matrices (AI HL)

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$

42 [Challenging] • Target: Integration using trigonometric identities • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$

b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

43 [Challenging] • Target: Integration using partial fractions • Cross-Lesson: 6.03 Markov Chains (AI HL)

Tabular integration (repeated parts): $\int x^3 e^x\,dx$. Number of IBP steps needed:

a. 2

b. 1

c. 4

d. 3

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44 [Challenging] • Target: Evaluating a definite integral • Cross-Lesson: 1.01 Proof by Mathematical Induction

Partial fractions for $\frac{1}{(x+1)^2}$ uses:

a. repeated linear factor

b. linear factors

c. irreducible quadratic

d. none

45 [Challenging] • Target: Integration involving inverse trigonometric results • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

$\int \sin^4 x\,dx$ using reduction formula:

a. $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

b. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

c. $\frac{x}{2} - \frac{\sin 4x}{8} + C$

d. $\frac{\sin^3 x \cos x}{4} + \frac{\sin x \cos x}{4} + \frac{x}{8} + C$

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